

Problem Set 2

Intermediate Macroeconomics, EC2211

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This problem set is based on material from LN3 and LN4. You are strongly encouraged to work through all questions before Seminar 2, but you should not hand in solutions. The quiz in Lecture 6 will test the same concepts and mechanisms.

1 Technology transfer in the Solow model

(Based on Exercise 2 in Jones, chapter 5) One explanation for China's rapid economic growth is the adoption of technologies previously developed in other countries. Policies that encourage international trade, foreign direct investment, and the diffusion of knowledge may facilitate such technology transfer.

Consider a Solow economy with

$$Y_t = A_t K_t^\alpha L^{1-\alpha}, \quad K_{t+1} = (1 - \delta)K_t + sY_t.$$

The labor force is constant, and the economy initially is in a steady state. At date $t = 0$, technology transfer raises the level of TFP by 20 percent. This is a one-time, permanent increase: after the jump, A_t remains constant at its new level.

- What happens immediately to capital per worker and output per worker?
- Show the change in a Solow diagram. What happens to investment at the old steady-state capital stock?
- Describe how capital and output per worker evolve during the transition. Compare their new steady-state levels with their old levels.
- What happens to the growth rate of output per worker when TFP increases, during the subsequent transition, and in the long run? Explain why the growth rate changes over time.
- Return to the economy's initial steady state. Instead of a one-time increase in the level of TFP, suppose that TFP begins to grow at the constant rate $g_A > 0$:

$$A_{t+1} = (1 + g_A)A_t.$$

What is the long-run growth rate of output per worker? Explain why this experiment has a different long-run implication from a one-time increase in the level of A .

2 Research productivity in the Romer model

(Based on Exercises 6 and 10 in Jones, chapter 6)

Consider the simplified Romer model used in Lecture 4:

$$\begin{aligned}Y_t &= A_t^\gamma L_{yt}, \\ \Delta A_{t+1} &= \zeta L_{at}, \\ L_{at} &= \lambda L_t, \\ L_{yt} &= (1 - \lambda)L_t, \\ L_{t+1} &= (1 + n)L_t.\end{aligned}$$

The economy initially is on a balanced growth path. In 2030, research productivity ζ falls immediately and permanently by 10 percent.

a. Use

$$g_{A,t} = \zeta \lambda \frac{L_t}{A_t}$$

to determine what happens immediately to the level and growth rate of knowledge, respectively.

- b. Describe how A_t/L_t and $g_{A,t}$ evolve after 2030. What is the long-run growth rate of knowledge?
- c. Draw the path of output per person, y_t , over time. Compare the new path with the path that would have prevailed without the decline in ζ .
- d. What happens to the growth rate and the level of output per person in the long run? Express the long-run growth rate in terms of the model parameters. Explain why a permanent decline in research productivity has a permanent level effect but only a temporary growth effect in this version of the Romer model.
- e. Instead suppose that the research share λ rises. What is the immediate cost of this change? What is its dynamic benefit? Why does a higher research share not necessarily increase output per person at every date?

3 From Malthus to innovation-driven growth

Answer each question briefly. Diagrams may be used where they help clarify the argument.

a. In the Malthusian model,

$$y_t = A_t \left(\frac{D}{L_t} \right)^\lambda, \quad \frac{L_{t+1}}{L_t} = \frac{y_t}{y^s}.$$

The economy initially is in a steady state. Analyze the short-run and long-run effects of a permanent increase in productivity A . What happens to income per person and population?

- b. Ideas are non-rival but may be partially excludable. Explain the difference between these two properties. Why can some excludability encourage the creation of ideas, and why can too much excludability be socially costly?
- c. Explain why price setting under perfect competition may not provide enough revenue to reward the costly creation of a non-rival idea. What role can temporary market power play?
- d. What is creative destruction? Explain why an innovation can increase aggregate productivity while reducing the value of incumbent firms and their technologies.
- e. Consider the claim: “Stronger patent protection and less competition necessarily lead to more innovation and faster economic growth.” Discuss the competing mechanisms emphasized in Lecture 4.
- f. Why might institutions that reduce total output nevertheless persist? Relate your answer to political power and the distribution of gains and losses.

Check your understanding

For each statement, decide whether it is true or false and explain your answer in one or two sentences.

1. A one-time permanent increase in the level of TFP permanently raises the growth rate of output per worker in the Solow model.
2. If TFP grows at rate g_A , output per worker grows at rate $g_A/(1 - \alpha)$ along the balanced growth path of the Solow model.
3. A permanent fall in research productivity ζ permanently lowers the long-run growth rate of output per person in the Romer model.
4. In the Romer model, a larger research share λ reduces current goods production but increases the production of new ideas.
5. In the Romer model, a permanent decline in population growth lowers the long-run growth rate of output per person.
6. Because patents provide incentives to innovate, stronger patent protection necessarily raises social welfare.

Recommended extra exercises

Exercises marked by a * are solved in the textbook.

- Jones chapter 5

- Exercise 5: Foreign aid
(Use the Solow model to analyze the effects of foreign aid in the form of physical capital.)
- Jones chapter 6:
 - Exercise 1: Nonrivalry
 - Exercise *2: Increasing returns and perfect competition
 - Exercise 8: The development of China and India
(Use the Romer model to analyze how increased research activity in China and India may affect GDP per capita in Europe and the United States through the global diffusion of ideas.)